

SWYNEX Technologies

Final Data Analytics Project – Employee Analytics

Task 4: Final Data Analytics Project

This case study combines data preparation, exploratory analysis, visualization, Power BI dashboarding and business insights using an employee dataset.

1. Problem Statement

Raw employee data can contain inconsistent categories and formats that make analysis difficult. The objective is to prepare the data, analyze workforce patterns and present the results through an interactive dashboard.

2. Dataset

The final dataset contains 1,020 employee records and 13 columns covering employee identity, age, department, region, status, joining date, salary, contact fields, performance score and remote-work status.

3. Data Quality

The supplied cleaned dataset contains 0 missing values and 0 duplicate rows. Performance categories were standardized to prevent separate 'poor' and 'Poor' groups, and Join_Date was parsed as a date field.

4. KPI Summary

Metric	Result
Total Employees	1,020
Average Salary	■85,164.30
Average Age	31.97
Remote Employees	513
Non-Remote Employees	507
Remote Work Share	50.29%

5. Key Findings

- DevOps has the largest workforce with 189 employees; Cloud Tech has 146.
- Sales has the highest departmental average salary at approximately ■86,874; Finance has the lowest at approximately ■82,280.
- California has the highest employee count among regions with 187; Texas has 153.
- Good is the largest performance category with 269 employees.
- Remote and non-remote employees are nearly evenly divided at 513 and 507.
- 2023 recorded the highest number of joiners (228) among 2020–2024; 2022 recorded the lowest (183).

6. Power BI Dashboard

The dashboard includes slicers for Department, Region, Performance Score, Remote Work and Age, along with KPI cards and visuals for region, department, remote work, performance, joining year and salary analysis.

7. Dashboard Validation

The visual titled 'Average salary by Department' should use Average aggregation for Salary. If it displays values around ■12M–■16M, it is using Sum rather than Average. The correct departmental averages are approximately ■82K–■87K.

8. Recommendations

Monitor workforce allocation by department, review salary differences, track performance distributions, monitor remote-work patterns, review regional workforce distribution and maintain consistent data categories and regular data-quality checks.

9. Conclusion

The project demonstrates an end-to-end data analytics workflow from cleaned data through EDA and Power BI dashboard development to business-focused insights and recommendations.